# Mikhail Razakov

## Compiler and Language Platform Engineer

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I develop UniversalToolchain/Wist2, a typed language composition and runtime SDK with cross-package plans, exact manifest binding, callable-first SSA, the public Wist facade, and experimental PlanFuzz.

| Typed composition plans | Callable-first SSA | 1,465 / 1,465 |
| --- | --- | --- |
| routes, passes, capabilities, exact executors | descriptor semantics and verifier-gated roundtrip | 0 failures; nine packages and clean consumers |

## EXPERIENCE

2024 — PRESENT

### UniversalToolchain / Wist2

- Typed composition/runtime SDK for independent language packages, deterministic routes/pass ordering, and schema-v5 lock identity.
- Fail-closed runtime policies, exact manifest binding, and lifecycle ownership.
- Callable-first SSA with descriptor semantics, effects, determinism, trust, and a verifier-gated roundtrip.
- Experimental PlanFuzz with a language-neutral core, an Acme adapter, a restricted Wist Int32 adapter, seven oracle families, replay, and reducers.

JUL–AUG 2026

### MCST · LLVM track

- C++23 graph-analysis core: RPO/cycles, Dijkstra, Tarjan SCC, and Dinic.
- CMake, warnings-as-errors, ASan/UBSan, clang-tidy, and I/O tests.

## SELECTED PROJECTS

### PlanFuzz

Experimental configuration-aware differential testing with a language-neutral core, Acme and restricted Wist Int32 adapters, fresh-process confirmation, and deterministic program/plan reduction.

### PS-form Analyzer

Ranked 1st of 49 with 104/104 tests; conservative memory-dependence analysis and an exact-oracle harness.

### x86-64 Codegen Lab

SSA-like IR, liveness, linear scan, iced-x86 emission, and differential validation.

## SKILLS

**Language architecture**
typed artifacts, contribution graphs, routes, passes, manifests

**IR / SSA**
Bytecode/AIR, callable descriptors, CFG, dominance, folding, SCCP-lite, DCE

**Verification**
differential/metamorphic testing, exact oracles, replay, reducers, parity

**Languages**
C#/.NET, C++23, C17, Python, Rust

## EDUCATION

Software Engineering student at HSE University

## RECOGNITION

HSE Vysshaya Proba prize-winner; top MEPhI Junior results in 2025/2026; Baltic Grand Prize.

## TECHNICAL COMMUNICATION

About 50 students; practical NASM IA-32 course; public documentation and technical writing.

## AVAILABILITY

Moscow / remote · up to 20 hours/week from September 2026